

project. Surround 360 comprises a 17-camera array, with 14 wide-angle cameras fixed on the flying-saucer-shaped surface of the camera, one fish-eye camera on top and two more on the bottom. This allows the device to capture the surroundings without being disturbed by the pole that holds it up.

Another innovation, according to the company, is that, cameras use a global shutter instead of a rolling one, which ensures the resulting footage does not display artefacts from the closing of individual shutters. The system also includes Web based software that captures and renders the images automatically.

There are several other premium 360-degree cameras such as Nokia's Ozo and Jaunt's Jaunt ONE. Cost-effective options are also available from companies like Nikon and GoPro.

If you want to take a really cheap shot at VR content, try something like Poppy, which is a purely optical device priced at around US\$ 30, which works with iPhones. Poppy uses mirrors to capture two stereographic images using your iPhone's camera. When you look through the viewfinder, Poppy lenses stitch the two images together into a single 3D view.

Closer home, Imagine is building Lenz, a platform that enables building and experiencing VR content without writing a single line of code. "Lenz enables 3D artists and 360 video makers to host their content and offer the experiences to the general public," says Satyanarayana. The Lenz platform is ready for internal evaluation and they are testing it with some of the top-tier corporations. With Lenz as the core platform, they are also developing different products for different use-cases by customising and changing the user experience as required.

Of course, the use of content is still low, unless it can be shared. Industry giants are focusing on that,

too. With existing VR platforms, it is quite easy for developers to create experiences, apps, games, etc using VR content and sell these through the relevant app stores. Videos can be shared through YouTube or Facebook.

In March, Google introduced VR View, which lets developers to easily embed 360-degree VR photos and videos on Android, iOS and the Web.

According to a blog post by Nathan Martz, product manager, Google, "For native apps, you can embed a VR view by grabbing the latest Cardboard SDK for Android or iOS and adding a few lines of code. On the Web, embedding a VR view is as simple as adding an iframe on your site. We are open-sourcing the HTML and JavaScript for Web developers on GitHub, so you can self-host and modify it to match your needs."

More to it than entertainment

When Facebook bought Oculus, Mark Zuckerberg wrote, "Imagine enjoying a courtside seat at a game, studying in a classroom of students and teachers all over the world or consulting with a doctor face-to-face, just by putting on goggles in your home."

Yes, it is these kinds of real applications that are going to make VR a part of everybody's life, not simply gaming or entertainment. The trend has already begun, in a big way, not just globally but in India, too.

At Facebook's annual developer conference, Zuckerberg spoke about many of their VR plans. In the future, people will be able to interact with things and places without actually buying the objects or travelling to the places. A US\$ 1 app, for example, could do the job of an expensive television!

Google is trying to take students places with their new Expeditions Pioneer Program. Schools, when selected for the programme, will be



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